



TCC805x MCUSP BSP

API Specification

for General Direct Memory Access Controller

Rev. 1.00 [A]

2020-11-12

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1 INTRODUCTION

This document describes the general direct memory access controller device driver (hereinafter referred to as GDMA driver) of TCC805x MCU BSP. For more detailed information of the direct memory access controller in the TCC805x MCU Sub-System, refer to Chapter 16 General Direct Memory Access (DMA) Controller of the “*PART 11-MICOM SUB-SYSTEM*” in the “*TCC805x Full Specification*”. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

DMA	Direct Memory Access
------------	----------------------

3 SOURCE FILES

3.1 Location of Source Files

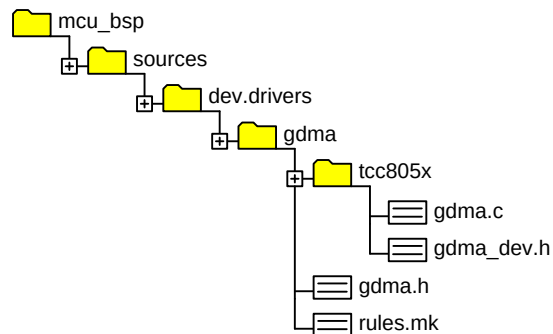


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- `gdma.c`
 - GDMA driver source
- `gdma_dev.h`
 - GDMA hardware definition header
- `gdma.h`
 - GDMA driver API header
- `rules.mk`
 - Make rules

4 CONSTANTS

4.1 GDMA_CH_MAX

Constant value for maximum GDMA channels

Declaration

```
#define GDMA_CH_MAX    (7UL)
```

Description

<i>Value</i>	<i>Description</i>
GDMA_CH_MAX	<i>The maximum number of channels</i>

Remark

None

5 STRUCTURES

5.1 GDMAControlData_t

Structure for controlling GDMA driver. You should initialize the channel number properly by setting the *cdChannel* variable.

Declaration

```
typedef struct _GDMAControlData_  
{  
    uint32 cdControl;    /* gdma control flags */  
    uint32 cdChannel;    /* control channel number for using */  
} GDMAControlData_t;
```

Description

Value	Description
<i>cdControl</i>	<i>Reserved</i>
<i>cdChannel</i>	<i>This value indicates the channel number for transferring the DMA request.</i>

Remark

None

6 APIs

6.1 GDMA_Initialize

Initializes the GDMA. There are two parameters in this function. The first parameter is for controlling the driver, and you should specify the channel index for this transfer using the "**cdChannel**" member variable of this first parameter. The second parameter is for changing the priority of ISR. If you want to use the default value, use **GIC_PRIORITY_NO_MEAN** as input value.

Declaration

```
sint32 GDMA_Initialize
(
    GDMAControlData_t *   pGdmaControl,
    uint32                 uiInterruptPriority
);
```

Parameters

Value	Description
<i>pGdmaControl</i>	[in] pointer of gdma control data. Refer to the GDMAControlData t.
<i>uiInterruptPriority</i>	[in] priority of gdma interrupt

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	Success
<i>SAL_RET_FAILED</i>	Failure

Remark

None

6.2 GDMA_TransferFunctionForM2M

Copies the data of source memory to destination memory. The first parameter is a pointer that points the control data, and the control data is passed variable when the driver was initialized.

Declaration

```
sint32 GDMA_TransferFunctionForM2M
(
    const GDMAControlData_t *    pGdmaControl,
    uint32                       uiSourceAddress,
    uint32                       uiDestinationAddress,
    uint32                       uiLength
);
```

Parameters

Value	Description
<i>pGdmaControl</i>	[in] pointer of GDMA control data. You should specify the channel index for this transfer using the “ cdChannel ” member variable of this structural parameter. Refer to the GDMAControlData t .
<i>uiSourceAddress</i>	[in] source address. This address is valid only for physical address.
<i>uiDestinationAddress</i>	[in] destination address. This address is valid only for physical address.
<i>uiLength</i>	[in] length of data requested for transfer

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	Success
<i>SAL_RET_FAILED</i>	Failure

Remark

None

7 HOW TO USE DMA DRIVER

In order to use the GDMA driver, the **GDMA_Initialize()** function must be called first. For initialization of GDMA driver, you should prepare the variable of **GDMAControlData_t** type and initialize the channel parameter(**cdChannel**).

The **GDMA_TransferFunctionForM2M()** function provides the memory to memory(M2M) copying function with blocking calling fashion. The meaning of blocking calling fashion is that the **GDMA_TransferFunctionForM2M()** function will not be returned until the transfer is completed. The **GDMA_TransferFunctionForM2M()** function introduces the OS-event-flag scheme for this blocking calling fashion. Therefore, the user task that calls this API will be pending in the suspend mode.

7.1 Operation of M2M (Memory-to-Memory) DMA Transfer

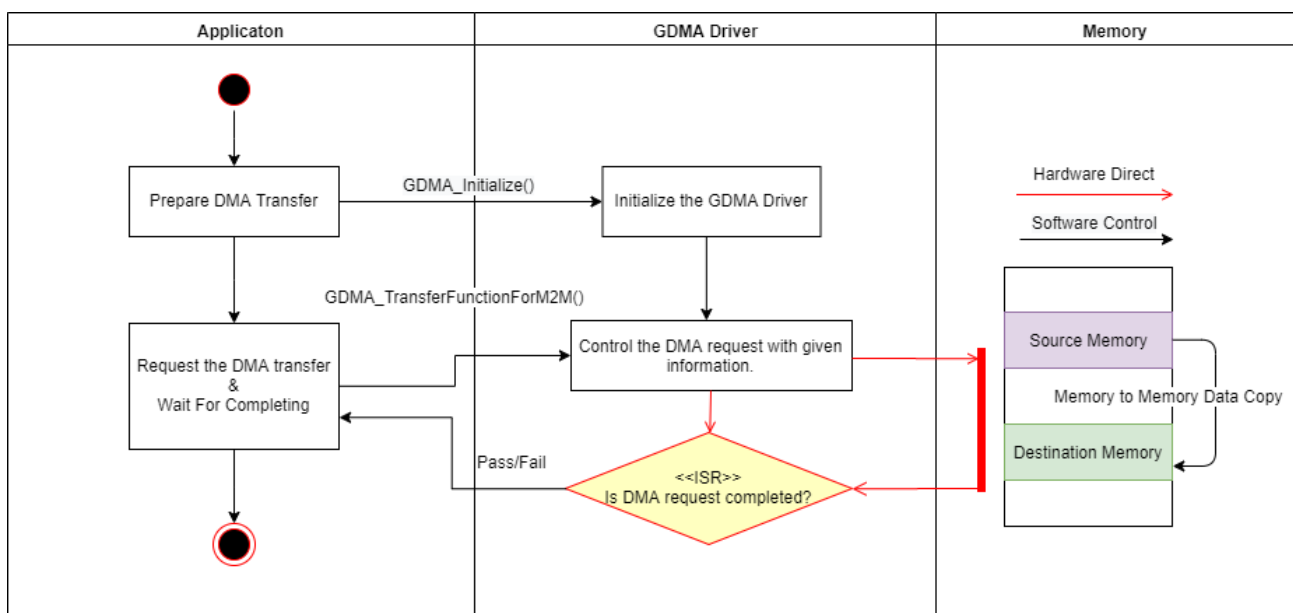


Figure 7.1 Activity Diagram for GDMA Driver

7.2 Initialize GDMA Driver

```

{
    GDMAControlData_t sample_pGdmaControl;
    sint32      ret;

    sample_pGdmaControl.cdChannel= 0;
    ret = GDMA_Initialize(&sample_pGdmaControl, GIC_PRIORITY_NO_MEAN);
}
  
```

7.3 Request M2M DMA Transfer

```
{  
    GDMAControlData_t sample_pGdmaControl;  
    uint32             request_size;  
    uint32             src_base;  
    uint32             dest_base;  
  
    request_size       = (1024 + (4*16) + 3);  
    src_base           = 0xC1000000;  
    dest_base          = (0xC1000000 + 0x1000);  
    sample_pGdmaControl.cdChannel= 0;  
  
    ret = GDMA_TransferFunctionForM2M(&sample_pGdmaControl, src_base, dest_base, request_size);  
}
```

8 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.00: 2020-11-12

- First version release

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